

Brownian free motion: the *velocitas ultima* denied, one action constant, and the floor

Result, 2026-09-28 (Claude; written derivation, to be refereed). A third route for the fifth-postulate programme, beside the deformation route (Theorems A, B) and the Gaussian state route (Theorem B’): the stochastic one. Let the free motion of a body of mass m be a Markov process with continuous paths, homogeneous in space and time and isotropic (premises P1–P3 of §1).

- **Theorem S1 (the law and the denial).** The free motion is Brownian motion with drift, with variance rate $D(m) \geq 0$ per coordinate. For $D = 0$ it is Newton’s inertial motion. For $D > 0$ the paths are almost surely nowhere differentiable, so the *velocitas ultima* that Newton’s scholium calls “certus & definitus” exists only as a mean velocity.
- **Theorem S2 (one constant).** If the law depends on the mass alone (P4) and the centre of mass of independent bodies obeys the law of their total mass (P5), then $\kappa = mD$ is one universal constant, with the dimension of an action; $\kappa = 0$ is admissible.
- **Theorem S3 (the floor).** For $\kappa > 0$, any record that decides from the path, with pinned ends, between the inertial chord and the constant-force parabola over a cell errs at equal priors with probability at least $\Phi(-\sqrt{K_\tau/2\kappa})$, attained by the optimal test, where $K_\tau = F^2\tau^3/(24m)$ is the Galileo action of the cell. Confidence $1 - \epsilon$ needs $K_\tau \geq 2z_{1-\epsilon}^2\kappa$; with $\kappa = \hbar$ this is the mark mesh of the Planck paper, eq. (2).
- **Proposition S4 (jumps allowed: the leap).** Dropping continuity, P2–P5 force the Lévy exponent of a body of mass m to be $m\Gamma(|k|/m)$ for one universal function Γ . If the only universal constant is an action κ , dimensional analysis forces $\Gamma(p) = C\kappa p^2$, and continuity follows. With a speed c as well, the relativistic kinetic energy divided by κ , $(\sqrt{m^2c^4 + c^2\kappa^2k^2} - mc^2)/\kappa$, is of this form and generates a process with jumps.
- **Independence (Nelson).** With diffusion coefficient $\nu = \kappa/2m$, Nelson’s stochastic mechanics obeys Newton’s second law in a mean sense and is equivalent to the Schrödinger equation with $\hbar = \kappa$ (Nelson 1966, abstract; precursor Fényes 1952, metadata). So every $\kappa \geq 0$ is consistent with the Laws, the analogue of Theorem A.

The route carries the four parts of the fifth-postulate goal: the statement (the existence of the ultimate velocity), its independence, exactly one added action constant, and a floor on recording the inertial–parabola difference, an area Newton takes to zero. Its limit is the same as in the other routes: $\kappa > 0$ is the negated postulate, assumed, and the zero branch stays admissible. Theorems S1 and S2 are elementary; the centre-of-mass argument of S2 is likely known in stochastic mechanics, where $\nu \propto 1/m$ is postulated, but it has not been located in print here.

1. Premises

- **P1 (continuity).** The free motion is a Markov process with continuous paths. Continuity of motion is Aristotle’s (*Physics* VI, cited by book only); the leap of al-Nazzam, already in the Planck paper’s classics, would be a jump, which P1 excludes.
- **P2 (homogeneity).** Stationary independent increments: no preferred place or time. A Galilean boost adds a constant drift and preserves P2.
- **P3 (isotropy).** No preferred direction.
- **P4 (mass alone).** A body’s free law depends only on its mass, as all bodies fall alike.
- **P5 (composition).** For independent bodies the centre of mass obeys the free law of the total mass.

2. Theorem S1

A process in $\mathbb{R}^3$ with stationary independent increments and continuous paths is a Brownian motion with drift, $X_t = x_0 + vt + \Sigma^{1/2}W_t$ (the Lévy–Itô decomposition has no jump part when the paths are continuous; standard). Isotropy gives $\Sigma = D I$, and $D \geq 0$ can depend only on the body. For $D > 0$, Brownian paths are almost surely nowhere differentiable (Paley, Wiener and Zygmund 1933, metadata), and they have Hausdorff dimension 2, the dimension found for quantum paths by Abbott and Wise (1981) (metadata). The difference quotient $(X_{t+h} - X_t)/h$ has no limit; its conditional mean does, $\lim_{h \downarrow 0} E[X_{t+h} - X_t | \mathcal{F}_t]/h = v$, which is Nelson’s mean forward velocity. $\square$

In the words of the scholium closing Book I, Section I (quoted from the 1687 text in §1 of the fifth-postulate note): “Extat limes quem velocitas in fine motus attingere potest, non autem transgredi. Hæc est velocitas ultima. [...] Cumq; hic limes sit certus & definitus”. For $D > 0$ the limit of the ratio exists for the mean and fails for the path. This is the denial of joint determinacy in the form Newton wrote it: the place at an instant is sharp, while the velocity at that instant has no pathwise value.

3. Theorem S2

Proof. Take independent bodies with masses m_i , drifts v_i and variance rates $D(m_i)$, and total mass $M = \sum m_i$. The centre of mass $X = \sum m_i x_i / M$ has the constant drift $\sum m_i v_i / M$ (momentum conservation) and the variance rate $\sum m_i^2 D(m_i) / M^2$. By P5 this equals $D(M)$, so the function $g(m) = m^2 D(m)$ satisfies $g(m_1 + m_2) = g(m_1) + g(m_2)$ for all $m_1, m_2 > 0$. Since $g \geq 0$, g is nondecreasing, and Cauchy’s equation gives $g(m) = \kappa m$. Hence $D(m) = \kappa / m$ and $mD = \kappa$ is the same for every body. Its dimension is mass times length²/time, an action. $\square$

The argument is the stochastic counterpart of the rotation-composition note, where interaction forces a common constant in the composition of rotations; P4 plays the role of the equivalence principle, and P5 the role of the centre-of-mass law. Neither argument excludes $\kappa = 0$.

4. Theorem S3

Let a constant force F act, so the body’s path is the Newtonian parabola plus $\sqrt{D}W_t$. Pin the ends of a cell of duration τ . The two laws to be told apart, with and without the force, are Brownian bridges of variance rate $D = \kappa/m$ around the parabola and around the chord. Their difference is the shift δx (chord minus parabola), with Cameron–Martin norm $\|\delta x\|^2 = D^{-1} \int_0^\tau \dot{\delta x}^2 dt = (m/\kappa) \int \dot{\delta x}^2 = 2K_\tau/\kappa$. The Neyman–Pearson test with equal priors errs with probability $\Phi(-\frac{1}{2}\|\delta x\|) = \Phi(-\sqrt{K_\tau/2\kappa})$, and no test does better. This is Proposition 6 of the cut-measure note with $\hbar$ replaced by κ ; its Theorem 2 distributes the evidence K_τ/κ over any sequence of cuts, each cut carrying the Kullback–Leibler share $3s(1-s)K_L/\kappa$. $\square$

With $\kappa = \hbar$, which is the normalization of the Euclidean free measure $e^{-S/\hbar}$ and of Nelson’s $\nu = \hbar/2m$, the threshold is $\tau \geq (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$, the Planck paper’s eq. (2).

5. Where the route stands

part of the goal	deformation route	state route	stochastic route
statement changed	commutativity (Thm A)	sharp states (Thm B')	<i>velocitas ultima</i> exists pathwise (S1)
independence from the Laws	Moyal (Thm A)	Gaussian closure	Nelson 1966
exactly one constant	Gutt, under covariance (Thm B)	Thm B'	composition, P4–P5 (S2)

part of the goal	deformation route	state route	stochastic route
floor on the comparison	Thms C, E	Thm C	Cameron–Martin (S3)

The premise that remains is the same in all three columns: why the constant is positive. In the stochastic route it has the plainest form, whether free paths have an instantaneous velocity, and Newton’s scholium asserts that they do. Two further limits: P4 and P5 are premises about bodies (an equivalence principle for fluctuations and a centre-of-mass law), and S3 is proved for additive position noise, the law of S1 with a force added; in Nelson’s full dynamics the drift depends on the state, and the corresponding bound is not claimed here. Allowing jumps (the leap) is treated in §5b.

5b. Jumps allowed (Proposition S4)

Drop P1 and keep P2–P5. The free law of a body of mass m is then an isotropic Lévy process; take it symmetric, so its exponent, $E e^{ik \cdot (X_t - X_0)} = e^{-t \phi(m, |k|)}$, is real and nonnegative. The centre of mass of independent bodies has exponent $\sum_i \phi(m_i, m_i |k|/M)$, and P5 requires $\phi(M, |k|) = \sum_i \phi(m_i, m_i |k|/M)$. Put $|k| = Mp$ and $G(m, p) = \phi(m, mp)$: then $G(m_1 + m_2, p) = G(m_1, p) + G(m_2, p)$, and $G \geq 0$ gives $G(m, p) = m \Gamma(p)$ with $\Gamma(p) = G(1, p)$. Hence

$$\phi(m, |k|) = m \Gamma(|k|/m), \quad \nu_m(A) = m \nu(mA),$$

where ν is the Lévy measure of Γ : a body of mass m jumps with the universal jumps shrunk by $1/m$ and their rate multiplied by m . Brownian motion is $\Gamma(p) = \kappa p^2/2$, giving $D = \kappa/m$ as in S2, and the symmetric α -stable laws are $\Gamma(p) = \kappa_\alpha p^\alpha$, with scale $\kappa_\alpha m^{1-\alpha}$. $\square$

Dimensional analysis. ϕ is an inverse time and $p = |k|/m$ has dimension $(\text{mass} \cdot \text{length})^{-1}$, so Γ has dimension $(\text{mass} \cdot \text{time})^{-1}$. If the only universal constant is one action κ , the unique monomial of that dimension is κp^2 , and no dimensionless combination of κ and p exists; so $\Gamma(p) = C \kappa p^2$ and the law is Brownian. Continuity is then a consequence: a jump needs a second constant. With a speed c available, $\kappa p/c$ is dimensionless and $\Gamma(p) = \kappa p^2 f(\kappa p/c)$ is allowed. The choice $\Gamma(p) = (\sqrt{c^4 + c^2 \kappa^2 p^2} - c^2)/\kappa$ gives $\phi = m \Gamma(|k|/m) = (\sqrt{m^2 c^4 + c^2 \kappa^2 k^2} - mc^2)/\kappa$, the relativistic kinetic energy divided by κ , which reduces to $\kappa k^2/2m$ for $\kappa |k| \ll mc$; it is a Bernstein function of k^2 , and its process has jumps (Carmona, Masters and Simon 1990, metadata). So in this reading the leap is what a speed limit readmits, and Aristotle’s continuity is what a single action constant enforces. This is a structural remark: which constants the free law may contain is itself a premise.

6. Consequence for STATE

Newton necessity gains a third route in which the negated postulate is Newton’s own sentence on the ultimate velocity: continuity, homogeneity and isotropy give Brownian free motion, composition gives one action constant, and the Cameron–Martin theorem gives the floor, equal to the Planck paper’s mark mesh at $\kappa = \hbar$. Positivity of κ remains a premise.